Incremental Predictive Value of Two Antenatal Visits for Recorded Low Birth Weight: Technical Report

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Generated by `python -m antenatal technical-report` on 2026-10-22. Every number is read from `outputs/`, `docs/evidence/` or `StudySpec`; none is typed by hand. The held out partition has not been evaluated, so all results are internal validation on development records.

1. Scope alignment

The proposal fixed the question, the cohort, the partition, the feature sets, the evaluation and a prototype before any model was fitted. The table maps each planned component to its status and to the evidence for it.

Planned feature	Purpose	Status	Evidence
Source data loader with checksum (Table 3.9)	Tie every result to one exact file	Completed	`src/antenatal/data/loader.py`
Cleaning and eligibility pipeline (R1)	Reproduce Table 3.5 and write a per record exclusion reason	Completed	`outputs/tables/cohort_flow.csv`, `outputs/cohort_manifest.csv`
Partition writer (R3)	Seal the held out partition before any model step	Completed	`partitions/holdout.sha256`
Feature construction (R2)	Build sets A, B and C with the availability screen	Completed	`src/antenatal/features/`
Nested cross validation module	Run the primary design on development records	Completed	`outputs/tables/feature_set_comparisons.csv`
Held out evaluation module (R5)	Evaluate once, logged with a timestamp	Partial: written and tested on synthetic data, not run	`src/antenatal/evaluation/holdout.py`
Model serialiser (R12, R16)	One artefact with checksum and version	Completed	`webapp/model/version.json`
Data model (Section 3.4.4)	Store records, visits, configurations and results	Completed in SQLite; the proposal budget named PostgreSQL	`src/antenatal/storage.py`, `python -m antenatal load-db`

Planned feature	Purpose	Status	Evidence
Sensitivity analyses S1 to S6	Test the comparison under changed rules	Completed	`outputs/tables/sensitivity_analyses.csv`
Interface screens 1 to 5 (Table 3.10)	Register, two visits, assessment, caseload	Completed as a single page over the API	`webapp/static/index.html`
Entry rules (R9, R10)	Ranges, timing and the synthetic flag	Completed	`docs/evidence/input_tests.md`
No clinical risk category (R11)	Rank and probability with interval only	Completed	`tests/integration/test_api.py`
Next.js interface, cross browser and performance tests	Production interface and evidence	Not implemented (Stage 6)	none yet
Container image	One command deployment	Not implemented: files written, never built	`Dockerfile`

2. Technical functionality and algorithms

Data and cohort. The released file (8,817 records) is read only after its SHA 256 matches the frozen value. Pure parsers turn text cells into typed values: blood pressure is read with a regular expression that takes the first pair of two or three digit numbers around a slash, a sentinel policy turns tokens such as "not_checked" into missing values, and a numeric parser repairs unambiguous slips such as a trailing full stop. Values outside the physiological ranges of `StudySpec` become missing and the record is kept. `CohortBuilder` then applies one named method per eligibility step and records the step and reason for every excluded record, giving the eligible cohort of 8,161 and the analysis cohort of 3,908 pregnancies with 206 low birth weight births.

Grouped stratified partitioning. Records that share a registration profile (age, last menstrual date, height, registration weight and prior deliveries) form a group. A stratified group split assigns whole groups to folds while balancing events; its first fold is the held out partition of 782 pregnancies with 41 events, written to file with a checksum before any screen or model step and readable only through a gate that logs every attempt. The development partition holds 3,126 pregnancies with 165 events.

Feature sets. Set A holds registration characteristics, set B adds the visit two measurements, and set C adds the visit one value of each measurement recorded at both visits for at least 60 percent of records. The availability screen never reads the outcome and is refitted in every training fold. For a linear model the change between visits is an exact linear combination of the two values, so set C adds values only; tree models also receive the change. Sets are subclasses of one abstract `FeatureSet`, so the runner never branches on a set's name.

Models. 8 families (L2 and L1 logistic regression, linear discriminant analysis, nearest neighbours, naive Bayes, a decision tree, a random forest and gradient boosted trees) and a constant prevalence baseline. Every

pipeline imputes medians with a missingness indicator and scales where needed, except the gradient boosted trees, which handle missing values natively. No class weighting or resampling is used, so probabilities stay on the natural scale.

Nested cross validation. 5 outer folds repeated 10 times, with hyperparameters chosen in 3 grouped inner folds by average precision. Outer test records take no part in any choice, and predictions are stored per record per repeat.

Paired group bootstrap. Predictions are averaged per pregnancy across repeats, the paired difference in a metric between two feature sets is computed on the same pregnancies, and registration groups are resampled 2,000 times to give a percentile interval. The bootstrap standard error gives the smallest detectable difference, (z for a two sided five percent test plus z for 80 percent power) times the standard error.

Interval on one probability. For the served L2 logistic model the coefficient covariance is the inverse of the penalised Hessian of the fitted design matrix. For a new case the standard error of the logit is computed from that covariance, and the Wald interval is transformed to the probability scale. It describes uncertainty in the estimate, not in the outcome.

Decision curve and retrieval. Net benefit counts true positives minus false positives weighted by the threshold odds. Retrieval reports the share of events in the top of the ranking against random ranking.

Data model. `python -m antenatal load-db` writes the research side of the proposal's data model into the application's SQLite file: pregnancy records with their eligibility and exclusion reason, cleaned visit measurements unique on record and visit number, the frozen model configurations, and every outer fold result unique on model, feature set, repeat and fold. The prototype's synthetic cases, visits and assessments sit in the same file. No table records which records are held out.

Prototype. A FastAPI application whose routers validate and route, whose services decide, and whose database stores synthetic cases, visits and assessments. Scoring goes only through the frozen artefact, whose checksum is verified when the application starts; a changed file stops the server. The interface shows rank, probability, the labelled interval, prevalence, event count, artefact version and a fixed limitation line, and never a risk category.

3. Testing approaches and results

The test suite ran 235 tests in 198.4 seconds with 0 failures: 188 unit tests, 41 integration tests and 6 browser tests. Line coverage of the package and the application was 97.6 percent of 2,910 statements. Ruff reported no findings and black reported no changes needed.

Approach	What it checks
Unit tests of parsers	Odd values from the file such as "121/20", "not_checked", "120 / 80", "36.5." and blanks
Cohort tests	Each count of Table 3.5, the eligible cohort, the sensitivity cohorts, that the raw frame is not modified
Partition tests	Group disjointness, stratification within one event, refusal to reseal, tamper detection, refusal of the held out partition without confirmation
Metric tests	Each metric against scikit-learn or a hand computation on toy data; bootstrap reproducibility; the power formula
Feature and model tests	Nested sets; no change terms in linear set C; no class weighting; native missing handling for the trees
Integration tests of the API	Every refusal path, a tampered artefact refusing to start, and equality of the API, the service and the package pipeline to 1e-9
End to end tests	The full command sequence on a throwaway project with a reduced specification
Input variation	12 inputs entered through a running server

The input variation run gave 12 of 12 inputs behaving as expected:

No.	Input	Expected	Observed status	Result
1	Typical complete case	201, assessment returned	201	pass
2	Synthetic flag not set	422, message contains "synthetic flag"	422	pass
3	Age 121	422, message contains "Maternal age must be between 12 and 55"	422	pass
4	Age 55, upper boundary	201, assessment returned	201	pass
5	Height missing	422, message contains "Height is required"	422	pass
6	Visit one systolic 300	422, message contains "Systolic blood pressure must be between 60 and 250"	422	pass
7	Visit one diastolic 20	422, message contains "Diastolic blood pressure must be between 30 and 160"	422	pass
8	Visit one at 30 weeks	422, message contains "at or before 28 weeks"	422	pass
9	Visit two at the same week as visit one	422, message contains "later than visit one"	422	pass
10	Visit two at 28 weeks, upper boundary	201, assessment returned	201	pass
11	Visit two pulse and haemoglobin missing	201, assessment returned	201	pass
12	Visit two entered before visit one	409, message contains "Record visit one before visit two"	409	pass

4. Code quality practices

The package separates data access, parsing, cohort rules, partitioning, features, models, evaluation, the artefact and reporting into their own modules, and every constant lives in one frozen `StudySpec`. Abstraction and polymorphism are used where there is more than one implementation: `FeatureSet` with its subclasses, and `MetricStrategy` with one subclass per metric. Every public class and function has a docstring and type hints. No function exceeds 50 lines and no file exceeds 300 lines, checked by a script. Ruff and black run locally and in the continuous integration workflow, every output file carries a checksum in `outputs/manifest.json`, and the README results and tree are generated and checked by a test. The history holds 64 commits, each with an imperative message stating what changed and why.

5. Results and limitations

Result	Value (95 percent interval)
Primary: first visit added (C over B), PR AUC, logistic	-0.0040 (-0.0130 to +0.0025)
Primary: first visit added (C over B), PR AUC, trees	-0.0066 (-0.0173 to +0.0024)
Smallest detectable difference, C over B, logistic	0.0110
Secondary: second visit added (B over A), ROC AUC, logistic	+0.0205 (-0.0066 to +0.0476)
Secondary: second visit added (B over A), ROC AUC, trees	+0.0345 (+0.0070 to +0.0627)

Result	Value (95 percent interval)
Sensitivity analyses with a C over B interval excluding zero	0 of 7
C minus B with all training data (learning curve, mean)	-0.0041
C over B, pooled multiple imputation	-0.0055 (-0.0151 to +0.0008)
Calibration slope, logistic, set C	0.98 (0.63 to 1.37)
Share of events in the top 10, 20 and 30 percent of the ranking	21.8 percent (15.9 to 27.9); 34.5 percent (27.8 to 42.0); 47.9 percent (40.4 to 55.4)
Pulse missing minus recorded, low birth weight rate	-2.5 points (-4.0 to -0.5)

Reading. Adding the first visit to the second did not improve prediction of recorded low birth weight by any amount this design could detect, in either model family, in any sensitivity analysis, under multiple imputation, or at any training size on the learning curve. The second visit over registration alone shows weak evidence of a gain in discrimination, with an interval excluding zero for the tree model but not for logistic regression. The predicted probabilities are calibrated within their uncertainty. Ranking by the model retrieves more low birth weight births than random ranking, which is a description of retrospective ranking only.

Limitations. The dataset release contains no ethics approval or consent statement; a request was sent to the producers on 21 October 2026. A newer version of the dataset exists on Zenodo, and this study stays on version 1 because the specification and every count were frozen against its checksum. It is one public Tanzanian dataset, with no facility or geographic identifier, so neither external nor geographic validation is possible. The development partition holds 165 events, so only differences of roughly the detectable size could be seen. Haemoglobin at visit two is recorded for a minority of pregnancies. The pulse missingness association reflects recording practice and cannot be separated from differences between clinics. The outcome is birth weight recorded under the care actually given, and nothing here shows that acting on a prediction would change it. The held out partition has not been evaluated, the Next.js interface and the container image are not built, and the prototype is a demonstration on synthetic cases that has not been validated for any pregnancy.